

Analytics for Decision Making

Q: Part A: Data Understanding & Cleaning

The original dataset contained 9,995 rows and was cleaned for consistency. No duplicates or missing values were found in the Frame Size column. Text fields were standardized using Text to Columns, and date fields were formatted consistently. Data quality checks identified date-format issues in *Order Date* and 7 inconsistent entries in *Ship Mode*, which were corrected using Text to Columns, category standardization, and the TRIM function. Postal Codes were validated using VLOOKUP, while other columns (Profit, Sales) required no changes.

Key checks carried out included:

- Missing values
- Duplicates (Data → Remove Duplicates)
- Inconsistent formatting (dates, currency, A-Z sorting , categorical entries) via Conditional Formatting and Home tab tools
- Blanks

Office Update To keep up-to-date with security updates, fixes, and improvements, choose Check for Updates. Check for Updates																		
State																		
Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name				
4/7/14	4/10/14	First Class	PI-19015	Pauline John	Consumer	United State	Mobile	Alabama	36608	South	FUR-FU-100	Furniture	Furnishings	Stacking Tray				
4/16/17	4/18/17	First Class	EB-13750	Edward Beck	Corporate	United State	Hoover	Alabama	35244	South	TEC-AC-1000	Technology	Accessories	SanDisk Ultra				
6/12/16	6/15/16	First Class	SC-20770	Stewart Carr	Corporate	United State	Decatur	Alabama	35601	South	OFF-AP-1000	Office Suppli	Appliances	1.7 Cubic Foot				
6/12/16	6/15/16	First Class	SC-20770	Stewart Carr	Corporate	United State	Decatur	Alabama	35601	South	OFF-BI-1000	Office Suppli	Binders	Avery Heavy-Duty				
11/26/16	11/29/16	First Class	MC-17425	Mark Cousin	Corporate	United State	Mobile	Alabama	36608	South	OFF-BI-1000	Office Suppli	Binders	Surelock Positioner				
11/26/16	11/29/16	First Class	MC-17425	Mark Cousin	Corporate	United State	Mobile	Alabama	36608	South	OFF-BI-1000	Office Suppli	Binders	Wilson Jones				
11/26/16	11/29/16	First Class	MC-17425	Mark Cousin	Corporate	United State	Mobile	Alabama	36608	South	TEC-MA-100	Technology	Machines	Hewlett-Packard				
11/26/16	11/29/16	First Class	MC-17425	Mark Cousin	Corporate	United State	Mobile	Alabama	36608	South	OFF-PA-1000	Office Suppli	Paper	Easy-staple 60				
4/16/17	4/18/17	First Class	EB-13750	Edward Beck	Corporate	United State	Hoover	Alabama	35244	South	OFF-SU-1000	Office Suppli	Supplies	Compact Auto				
12/20/14	12/22/14	First Class	CT-11995	Carol Triggs	Consumer	United State	Tucson	Arizona	85705	West	TEC-AC-1000	Technology	Accessories	Logitech 910				
11/6/15	11/9/15	First Class	MT-17815	Meg Tillman	Consumer	United State	Scottsdale	Arizona	85254	West	OFF-BI-1000	Office Suppli	Binders	UniKeep View				
7/19/15	7/20/15	First Class	TS-21085	Thais Sissma	Consumer	United State	Phoenix	Arizona	85023	West	OFF-BI-1000	Office Suppli	Binders	Wilson Jones				
10/2/16	10/3/16	First Class	LP-17095	Liz Preis	Consumer	United State	Phoenix	Arizona	85023	West	OFF-BI-1000	Office Suppli	Binders	Premier Ellip				
1/19/14	1/20/14	First Class	TS-21340	Toby Swindle	Consumer	United State	Scottsdale	Arizona	85254	West	OFF-BI-1000	Office Suppli	Binders	GBS Standard				
1/19/14	1/20/14	First Class	TS-21340	Toby Swindle	Consumer	United State	Scottsdale	Arizona	85254	West	FUR-BO-100	Furniture	Bookcases	O'Sullivan 4-				
12/20/14	12/22/14	First Class	CT-11995	Carol Triggs	Consumer	United State	Tucson	Arizona	85705	West	FUR-CH-100	Furniture	Chairs	Global Low E				
11/6/15	11/9/15	First Class	MT-17815	Meg Tillman	Consumer	United State	Scottsdale	Arizona	85254	West	OFF-EN-1000	Office Suppli	Envelopes	Tydel Strip #1				
1/19/14	1/20/14	First Class	TS-21340	Toby Swindle	Consumer	United State	Scottsdale	Arizona	85254	West	OFF-EN-1000	Office Suppli	Envelopes	Tyvek Top-O				
12/20/14	12/22/14	First Class	CT-11995	Carol Triggs	Consumer	United State	Tucson	Arizona	85705	West	FUR-FU-100	Furniture	Furnishings	Computer R				
12/20/14	12/22/14	First Class	CT-11995	Carol Triggs	Consumer	United State	Tucson	Arizona	85705	West	OFF-PA-1000	Office Suppli	Paper	Xerox 1888				
12/20/14	12/22/14	First Class	CT-11995	Carol Triggs	Consumer	United State	Tucson	Arizona	85705	West	OFF-PA-1000	Office Suppli	Paper	Xerox 1954				
1/19/14	1/20/14	First Class	TS-21340	Toby Swindle	Consumer	United State	Scottsdale	Arizona	85254	West	OFF-PA-1000	Office Suppli	Paper	TOPS Carbon				
4/24/17	4/25/17	First Class	SV-20935	Susan Vittori	Consumer	United State	Mesa	Arizona	85204	West	TEC-PH-1000	Technology	Phones	AT&T 84100				
7/26/15	7/28/15	First Class	CA-12775	Cynthia Arnt	Consumer	United State	Mesa	Arizona	85204	West	FUR-TA-100	Furniture	Tables	Bevis Boat-S				
11/10/17	11/11/17	First Class	AB-10600	Ann Blume	Corporate	United State	Tucson	Arizona	85705	West	TEC-AC-1000	Technology	Accessories	Case Logic 2				
10/2/14	10/5/14	First Class	ES-14080	Erin Smith	Corporate	United State	Tempe	Arizona	85281	West	TEC-AC-1000	Technology	Accessories	Kingston Dig				
9/11/16	9/13/16	First Class	LC-16930	Linda Cazam	Corporate	United State	Phoenix	Arizona	85023	West	OFF-AR-1000	Office Suppli	Art	Crayola Anti				
11/10/17	11/11/17	First Class	AB-10600	Ann Blume	Corporate	United State	Tucson	Arizona	85705	West	OFF-BI-1000	Office Suppli	Binders	Cardinal Easy				
5/11/17	5/13/17	First Class	TS-21370	Todd Sumral	Corporate	United State	Phoenix	Arizona	85023	West	FUR-BO-100	Furniture	Bookcases	Bestar Classi				
11/10/17	11/11/17	First Class	AB-10600	Ann Blume	Corporate	United State	Tucson	Arizona	85705	West	FUR-CH-100	Furniture	Chairs	Hon 2090 IPI				
10/2/14	10/5/14	First Class	ES-14080	Erin Smith	Corporate	United State	Tempe	Arizona	85281	West	OFF-EN-1000	Office Suppli	Envelopes	Letter or Leg				
10/2/14	10/5/14	First Class	ES-14080	Erin Smith	Corporate	United State	Tempe	Arizona	85281	West	OFF-FA-1000	Office Suppli	Fasteners	Staples				
11/10/17	11/11/17	First Class	AB-10600	Ann Blume	Corporate	United State	Tucson	Arizona	85705	West	TEC-MA-100	Technology	Machines	DYMO CardS				
10/2/14	10/5/14	First Class	ES-14080	Erin Smith	Corporate	United State	Tempe	Arizona	85281	West	OFF-PA-1000	Office Suppli	Paper	Adams Telep				
9/11/16	9/13/16	First Class	LC-16930	Linda Cazam	Corporate	United State	Phoenix	Arizona	85023	West	OFF-PA-1000	Office Suppli	Paper	Xerox 1976				
9/11/16	9/13/16	First Class	LC-16930	Linda Cazam	Corporate	United State	Phoenix	Arizona	85023	West	OFF-PA-1000	Office Suppli	Paper	Great White				
7/14/16	7/17/16	First Class	ES-14080	Erin Smith	Corporate	United State	Tucson	Arizona	85705	West	TEC-PH-1000	Technology	Phones	LG Electronic				
11/10/17	11/11/17	First Class	AB-10600	Ann Blume	Corporate	United State	Tucson	Arizona	85705	West	TEC-PH-1000	Technology	Phones	Adtran 1202				

Analytics for Decision Making

Office Update																	
To keep up-to-date with security updates, fixes, and improvements, choose Check for Updates.																	
Category																	
E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	
Mod	Customer	Customer	Segment	Country	City	State	Postal	Region	Product	Category	Sub-Category	Product	Sales				
nd Class EH-13945	Eric Hoffman	Consumer	United State	Los Angeles	California	90049	West	TEC-AC-1000	Technology	Accessories	Imation18GE	90.57					
Class OH-18715	Odella Nelson	Corporate	United State	Eagan	Minnesota	55122	Central	TEC-AC-1000	Technology	Accessories	Verbatim25	45.98					
ard Cla LH-16900	Lena Herman	Consumer	United State	Dover	Delaware	19901	East	TEC-AC-1000	Technology	Accessories	Imation18Gb	45					
Class TB-21055	Ted Butterfli	Consumer	United State	Troy	New York	12180	East	TEC-AC-1000	Technology	Accessories	Imation18Gb	30					
ard Cla KM-16720	Kunst Miller	Consumer	United State	Los Angeles	California	90004	West	TEC-AC-1000	Technology	Accessories	Verbatim25	13.98					
ard Cla PG-18895	Paul Gonzale	Consumer	United State	Rochester	Minnesota	55901	Central	TEC-AC-1000	Technology	Accessories	Logitech152	19.99					
ard Cla RB-19465	Rick Bensley	Home Office	United State	Chicago	Illinois	60610	Central	TEC-AC-1000	Technology	Accessories	Sandisk Ultra	95.976					
ard Cla LC-16870	Lena Caciop	Consumer	United State	Aurora	Colorado	80013	West	TEC-AC-1000	Technology	Accessories	Logitech K35	238.896					
ard Cla JM-15250	Janet Martin	Consumer	United State	Charlotte	North Caroli	28205	South	TEC-AC-1000	Technology	Accessories	Memorex MI	74.112					
ard Cla PA-19060	Pete Armstr	Home Office	United State	Orland Park	Illinois	60462	Central	TEC-AC-1000	Technology	Accessories	Logitech1Ga	339.96					
nd Class EM-13960	Eric Murdock	Consumer	United State	Philadelphia	Pennsylvania	19134	East	TEC-AC-1000	Technology	Accessories	Lenovo 17-K6	54.384					
nd Class DK-13150	David Kendri	Corporate	United State	Decatur	Illinois	62521	Central	TEC-AC-1000	Technology	Accessories	Imation1Sec	408.744					
nd Class RM-19675	Robert Marli	Home Office	United State	Monroe	Louisiana	71203	South	TEC-AC-1000	Technology	Accessories	HP Standard	29					
ard Cla FM-14290	Frank Mervl	Home Office	United State	Los Angeles	California	90032	West	TEC-AC-1000	Technology	Accessories	Sandisk Ultra	176.8					
nd Class MS-17980	Michael Stev	Corporate	United State	Dallas	Texas	75220	Central	TEC-AC-1000	Technology	Accessories	Memorex MI	20.8					
ard Cla VW-21775	Victoria Wils	Corporate	United State	Medina	Ohio	44256	East	TEC-AC-1000	Technology	Accessories	Sony Micro V	89.584					
ard Cla CS-12400	Christopher J	Home Office	United State	Tampa	Florida	33614	South	TEC-AC-1000	Technology	Accessories	Logitech1P7	617.976					
ard Cla KC-16540	Kelly Colliste	Consumer	United State	San Diego	California	92037	West	TEC-AC-1000	Technology	Accessories	Logitech1P7	3347.37					
nd Class CC-12670	Craig Carrel	Consumer	United State	New York Cit	New York	10009	East	TEC-AC-1000	Technology	Accessories	Sabrent 4-Pc	20.37					
ard Cla AH-10210	Alan Hwang	Consumer	United State	Brentwood	California	94513	West	TEC-AC-1000	Technology	Accessories	Microsoft Sc	79.9					
Class HA-14920	Harold Hale	Corporate	United State	Richardson	Texas	75080	Central	TEC-AC-1000	Technology	Accessories	Logitech 910	209.93					
ard Cla TB-21520	Tracy Blumst	Consumer	United State	Portland	Oregon	97206	West	TEC-AC-1000	Technology	Accessories	Microsoft An	239.8					
nd Class NZ-18565	Nick Zandus	Home Office	United State	Columbus	Ohio	43229	East	TEC-AC-1000	Technology	Accessories	Logitech Wir	255.984					
ard Cla NM-18445	Nathan Mau	Home Office	United State	New York Cit	New York	10024	East	TEC-AC-1000	Technology	Accessories	NETGEAR N	360					
ard Cla GK-14620	Grace Kelly	Corporate	United State	Hesperia	California	92345	West	TEC-AC-1000	Technology	Accessories	Logitech Wir	99.99					
Class AB-10060	Adam Bellav	Home Office	United State	New York Cit	New York	10009	East	TEC-AC-1000	Technology	Accessories	Sabrent 4-Pc	6.79					
ard Cla AG-10900	Arthur Gainc	Consumer	United State	Tucson	Arizona	85705	West	TEC-AC-1000	Technology	Accessories	NETGEAR AC	1023.936					
ard Cla AG-10900	Arthur Gainc	Consumer	United State	Tucson	Arizona	85705	West	TEC-AC-1000	Technology	Accessories	Microsoft Na	479.04					
nd Class LA-16780	Laura Armst	Corporate	United State	Taylor	Michigan	48180	Central	TEC-AC-1000	Technology	Accessories	Sony 32GB G	41.9					
ard Cla DO-13435	Denny Ordw	Consumer	United State	Pembroke Pl	Florida	33024	South	TEC-AC-1000	Technology	Accessories	Logitech Mex	83.976					
ard Cla MV-18190	Mike Vittori	Consumer	United State	New York Cit	New York	10009	East	TEC-AC-1000	Technology	Accessories	Logitech152	39.98					
nd Class AG-10675	Anna Gaiyma	Consumer	United State	Houston	Texas	77036	Central	TEC-AC-1000	Technology	Accessories	Sony 64GB C	57.584					
ard Cla KH-16510	Keith Herre	Consumer	United State	San Francisco	California	94110	West	TEC-AC-1000	Technology	Accessories	Sony 64GB C	179.95					
Class JE-15745	Joel Eaton	Consumer	United State	Louisville	Colorado	80027	West	TEC-AC-1000	Technology	Accessories	KeyTronic16	196.752					
ard Cla RB-19795	Ross Baird	Home Office	United State	Gastonia	North Caroli	28052	South	TEC-AC-1000	Technology	Accessories	Imation1Sec	408.744					
ard Cla RB-19795	Ross Baird	Home Office	United State	Gastonia	North Caroli	28052	South	TEC-AC-1000	Technology	Accessories	Imation1Sec	291.96					

Home

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Conditional Formatting

Format as Table

Cell Styles

Cells

Editing

Create PDF and Share Link

Office Update

To keep up-to-date with security updates, fixes, and im

Date

Updates

Check for Updates

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	
ID	Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer	Segment	Country	City	State	Postal Code	Region	Product Line	Category	Sub-Category		
1	CA-2016-152	11/8/16	11/11/16	11/11/16	Standard Class CG-12520	Claire Gute				Kentucky	42420	South	FUR-BO-1000 Furniture	Bookcases			
2	CA-2016-152	11/8/16	11/11/16	11/11/16	Standard Class CG-12520	Claire Gute				Kentucky	42420	South	FUR-CH-1000 Furniture	Chairs			
3	CA-2016-158	6/12/16	6/16/16	6/16/16	Standard Class DV-13045	Darin Van				California	90036	West	OFF-LA-1000 Office Supply	Labels			
4	US-2015-108	10/11/15	10/18/15	10/18/15	Standard Class SO-20335	Sean O'Doe				Florida	33311	South	OFF-ST-1000 Office Supply	Storage			
6	CA-2014-115	6/9/14	6/14/14	6/14/14	Standard Class BH-11710	Brosina Ho				California	90032	West	FUR-FU-1000 Furniture	Furniture			
7	CA-2014-115	6/9/14	6/14/14	6/14/14	Standard Class BH-11710	Brosina Ho				California	90032	West	OFF-AR-1000 Office Supply	Art			
8	CA-2014-115	6/9/14	6/14/14	6/14/14	Standard Class BH-11710	Brosina Ho				California	90032	West	TEC-PH-1000 Technology	Phones			
9	CA-2014-115	6/9/14	6/14/14	6/14/14	Standard Class BH-11710	Brosina Ho				California	90032	West	OFF-BI-1000 Office Supply	Binders			
10	CA-2014-115	6/9/14	6/14/14	6/14/14	Standard Class BH-11710	Brosina Ho				California	90032	West	OFF-AP-1000 Office Supply	Appliances			
11	CA-2014-115	6/9/14	6/14/14	6/14/14	Standard Class BH-11710	Brosina Ho				California	90032	West	FUR-TA-1000 Furniture	Tables			
12	CA-2014-115	6/9/14	6/14/14	6/14/14	Standard Class BH-11710	Brosina Ho				California	90032	West	TEC-PH-1000 Technology	Phones			
13	CA-2017-114	4/15/17	4/20/17	4/20/17	Standard Class AA-10480	Andrew All				North Carolina	28027	South	OFF-PA-1000 Office Supply	Paper			
14	CA-2016-162	12/5/16	12/10/16	12/10/16	Standard Class IM-15070	Irene Made				Washington	98103	West	OFF-BI-1000 Office Supply	Binders			
15	US-2015-118	11/22/15	11/26/15	11/26/15	Standard Class HP-14815	Harold Pav				Texas	76106	Central	OFF-AP-1000 Office Supply	Appliances			
16	US-2015-118	11/22/15	11/26/15	11/26/15	Standard Class HP-14815	Harold Pav				Texas	76106	Central	OFF-BI-1000 Office Supply	Binders			
17	CA-2014-105	11/11/14	11/18/14	11/18/14	Standard Class PK-19075	Pete Kitz				Wisconsin	53711	Central	OFF-ST-1000 Office Supply	Storage			
18	CA-2014-165	5/11/14	5/15/14	5/15/14	Second Class AG-10270	Alejandro C				Utah	84084	West	OFF-ST-1000 Office Supply	Storage			
19	CA-2014-144	8/27/14	9/1/14	9/1/14	Second Class ZD-21925	Zuschuss D				California	94109	West	OFF-AR-1000 Office Supply	Art			
20	CA-2014-144	8/27/14	9/1/14	9/1/14	Second Class ZD-21925	Zuschuss D				California	94109	West	TEC-PH-1000 Technology	Phones			
21	CA-2014-144	8/27/14	9/1/14	9/1/14	Second Class ZD-21925	Zuschuss D				California	94109	West	OFF-BI-1000 Office Supply	Binders			
22	CA-2016-139	12/9/16	12/13/16	12/13/16	Standard Class RB-16585	Ken Black				Nebraska	68025	Central	OFF-AR-1000 Office Supply	Art			
23	CA-2016-139	12/9/16	12/13/16	12/13/16	Standard Class RB-16585	Ken Black				Nebraska	68025	Central	OFF-AP-1000 Office Supply	Appliances			
24	US-2017-156	7/16/17	7/18/17	7/18/17	Second Class SF-20065	Sandra Fla				Pennsylvania	19140	East	FUR-CH-1000 Furniture	Chairs			
25	CA-2015-106	9/25/15	9/30/15	9/30/15	Standard Class EB-13870	Emily Burns				United States	Greenville	South Carolina	29615	South	FUR-TA-1000 Furniture	Tables	
26	CA-2016-122	1/16/16	1/20/16	1/20/16	Second Class EH-13945	Eric Hoffman				United States	Los Angeles	California	90049	West	OFF-BI-1000 Office Supply	Binders	
27	CA-2016-122	1/16/16	1/20/16	1/20/16	Second Class EH-13945	Eric Hoffman				United States	Los Angeles	California	90049	West	TEC-AC-1000 Technology	Accessories	
28	US-2015-156	9/17/15	9/21/15	9/21/15	Standard Class TB-21520	Tracy Blumst				United States	Philadelphia	Pennsylvania	19140	East	FUR-BO-1000 Furniture	Bookcases	
29	US-2015-156	9/17/15	9/21/15	9/21/15	Standard Class TB-21520	Tracy Blumst				United States	Philadelphia	Pennsylvania	19140	East	OFF-BI-1000 Office Supply	Binders	
30	US-2015-156	9/17/15	9/21/15	9/21/15	Standard Class TB-21520	Tracy Blumst				United States	Philadelphia	Pennsylvania	19140	East	FUR-FU-1000 Furniture	Furniture	
31	US-2015-156	9/17/15	9/21/15	9/21/15	Standard Class TB-21520	Tracy Blumst				United States	Philadelphia	Pennsylvania	19140	East	OFF-EN-1000 Office Supply	Envelopes	
32	US-2015-156	9/17/15	9/21/15	9/21/15	Standard Class TB-21520	Tracy Blumst				United States	Philadelphia	Pennsylvania	19140	East	OFF-AR-1000 Office Supply	Art	
33	US-2015-156	9/17/15	9/21/15	9/21/15	Standard Class TB-21520	Tracy Blumst				United States	Philadelphia	Pennsylvania	19140	East	OFF-BI-1000 Office Supply	Binders	
34	US-2015-156	9/17/15	9/21/15	9/21/15	Standard Class TB-21520	Tracy Blumst				United States	Philadelphia	Pennsylvania	19140	East	OFF-AR-1000 Office Supply	Art	
35	CA-2017-107	10/19/17	10/23/17	10/23/17	Second Class MA-17560	Matt Abelm				United States	Houston	Texas	77095	Central	OFF-PA-1000 Office Supply	Paper	
36	CA-2016-117	12/8/16	12/10/16	12/10/16	First Class GH-14485	Gene Hale				United States	Richardson	Texas	75080	Central	TEC-PH-1000 Technology	Phones	
37	CA-2016-117	12/8/16	12/10/16	12/10/16	First Class GH-14485	Gene Hale				United States	Richardson	Texas	75080	Central	FUR-FU-1000 Furniture	Furniture	
38	CA-2015-117	12/27/15	12/31/15	12/31/15	Standard Class SN-20710	Steve News				United States	Houston	Texas	77091	Central	OFF-EN-1000 Office Supply	Envelopes	

General

No specific format

Number

Order Date

Currency

Order Date

Accounting

Order Date

Short Date

Order Date

Long Date

Order Date

Time

Order Date

Percentage

Order Date

Fraction

Order Date

Scientific

Order Date

Text

ABC Order Date

More Number Formats...

Sample - Supertore 4

+

Analytics for Decision Making

Superstore Data Clean Record:

These calculations were applied across all rows, and Profit Margin values were formatted as percentages using the home tab's Number formatting tools. The final dataset achieved 99.12% data integrity, ensuring a reliable foundation for business analysis with standardized formats and fully validated calculations.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit		
1	541	CA-2014-	2/1/2014	2/3/2014	First Class	BD-11500	Bradley Drucker	Consumer	United States	Green Bay	Wisconsin	54302	Central	TEC-AC-1	Technology Accessories	Enermax Aurora	\$468.90	6	\$0.00	\$206.32	
2	676	CA-2017-	12/5/2017	12/8/2017	First Class	RB-19570	Rob Beeghly	Consumer	United States	Columbus	Indiana	47201	Central	TEC-AC-1	Technology Accessories	Imation 1608M	\$99.39	3	\$0.00	\$40.75	
3	809	CA-2015-	2/3/2015	2/5/2015	First Class	AA-10375	Allen Arnold	Consumer	United States	Omaha	Nebraska	68104	Central	TEC-AC-1	Technology Accessories	Kensington Slim	\$149.97	3	\$0.00	\$50.99	
4	1567	US-2015-	11/29/2015	11/30/2015	First Class	AW-1084	Anthony Witt	Consumer	United States	Allen	Texas	75002	Central	TEC-AC-1	Technology Accessories	Kingston Digital	\$21.48	3	\$0.20	\$0.27	
5	6385	US-2017-	1/15/2017	1/18/2017	First Class	TB-21250	Tim Brockman	Consumer	United States	Austin	Texas	78745	Central	TEC-AC-1	Technology Accessories	Logitech 910-01	\$47.98	2	\$0.20	\$14.40	
6	7652	CA-2017-	8/7/2017	8/8/2017	First Class	CK-12205	Chloris Kastens	Consumer	United States	Dallas	Texas	75081	Central	TEC-AC-1	Technology Accessories	Logitech K350	\$119.45	3	\$0.20	\$13.44	
7	2058	CA-2017-	12/22/2017	12/25/2017	First Class	TP-21130	Theone Pipping	Consumer	United States	Detroit	Michigan	48227	Central	TEC-AC-1	Technology Accessories	Logitech Gamin	\$84.99	1	\$0.00	\$30.60	
8	8509	CA-2015-	12/11/2015	12/14/2015	First Class	CA-12775	Cynthia Arntzen	Consumer	United States	Detroit	Michigan	48205	Central	TEC-AC-1	Technology Accessories	Maxell 4.7GB DV	\$175.23	11	\$0.00	\$61.33	
9	6386	US-2017-	1/15/2017	1/18/2017	First Class	TB-21250	Tim Brockman	Consumer	United States	Austin	Texas	78745	Central	TEC-AC-1	Technology Accessories	Maxell 74 Minut	\$62.59	8	\$0.20	\$13.30	
10	2056	CA-2017-	12/22/2017	12/25/2017	First Class	TP-21130	Theone Pipping	Consumer	United States	Detroit	Michigan	48227	Central	TEC-AC-1	Technology Accessories	Microsoft Wirel	\$199.95	5	\$0.00	\$63.98	
11	512	CA-2017-	11/26/2017	11/27/2017	First Class	LS-17245	Lynn Smith	Consumer	United States	Gladstone	Missouri	64118	Central	TEC-AC-1	Technology Accessories	SanDisk Cruzer	\$38.04	2	\$0.00	\$12.17	
12	9624	CA-2017-	6/29/2017	6/30/2017	First Class	ME-17725	Max Engle	Consumer	United States	Dallas	Texas	75220	Central	OFF-AP-1	Office Supp	Appliances	Belkin F90930V	\$21.39	2	\$0.80	-\$54.55
13	8020	CA-2017-	11/2/2017	11/5/2017	First Class	NP-18670	Nora Paige	Consumer	United States	Saint Louis	Missouri	63116	Central	OFF-AP-1	Office Supp	Appliances	Black & Decker	\$83.90	10	\$0.00	\$20.98
14	6990	CA-2017-	12/11/2017	12/13/2017	First Class	DK-13375	Dennis Kane	Consumer	United States	Ablene	Texas	79605	Central	OFF-AP-1	Office Supp	Appliances	Hoover Comme	\$1.39	2	\$0.80	-\$3.76
15	674	CA-2017-	12/5/2017	12/8/2017	First Class	RB-19570	Rob Beeghly	Consumer	United States	Columbus	Indiana	47201	Central	OFF-AP-1	Office Supp	Appliances	Kensington 6.0	\$61.44	3	\$0.00	\$16.59
16	6252	CA-2014-	12/2/2014	12/4/2014	First Class	MC-17575	Matt Collins	Consumer	United States	Chicago	Illinois	60623	Central	OFF-AP-1	Office Supp	Appliances	Staple holder	\$2.39	1	\$0.80	-\$6.34
17	5310	CA-2015-	11/19/2017	11/21/2017	First Class	NC-18415	Nathan Cano	Consumer	United States	Houston	Texas	77095	Central	OFF-AR-1	Office Supp	Art	Avery Hi-Liter Gl	\$13.04	5	\$0.20	\$3.91
18	2690	CA-2014-	6/30/2014	7/2/2014	First Class	RA-19915	Russell Appleby	Consumer	United States	Chicago	Illinois	60653	Central	OFF-AR-1	Office Supp	Art	BIC Brite Liner G	\$5.25	4	\$0.20	\$1.64
19	8321	CA-2015-	12/5/2015	12/6/2015	First Class	SF-20065	Sandra Flanagan	Consumer	United States	Dallas	Texas	75220	Central	OFF-AR-1	Office Supp	Art	Boston Electric	\$45.04	2	\$0.20	\$4.50
20	7296	CA-2014-	5/4/2014	5/5/2014	First Class	NC-1834C	Nat Carroll	Consumer	United States	Houston	Texas	77036	Central	OFF-AR-1	Office Supp	Art	Boston Home &	\$37.84	2	\$0.20	\$2.84
21	3901	CA-2016-	3/14/2016	3/17/2016	First Class	JS-15880	John Stevenson	Consumer	United States	Harlingen	Texas	78550	Central	OFF-AR-1	Office Supp	Art	BOSTON Range	\$145.54	7	\$0.20	\$16.37
22	2918	CA-2017-	8/27/2017	8/30/2017	First Class	SE-20110	Sanjit Engle	Consumer	United States	Dallas	Texas	75220	Central	OFF-AR-1	Office Supp	Art	Eberhard Faber	\$5.95	1	\$0.20	\$0.37
23	1793	CA-2014-	12/1/2014	12/3/2014	First Class	RP-19390	Resi Polking	Consumer	United States	Madison	Wisconsin	53711	Central	OFF-AR-1	Office Supp	Art	Hunt BOSTON V	\$46.64	4	\$0.00	\$12.59
24	6409	CA-2017-	5/14/2017	5/15/2017	First Class	GT-14710	Greg Tran	Consumer	United States	Houston	Texas	77041	Central	OFF-AR-1	Office Supp	Art	Newell 309	\$46.20	5	\$0.20	\$5.78
25	2823	CA-2017-	9/18/2017	9/20/2017	First Class	DC-1285C	Dan Campbell	Consumer	United States	Arlington	Texas	76017	Central	OFF-AR-1	Office Supp	Art	Newell 314	\$8.93	2	\$0.20	\$0.56
26	3725	CA-2017-	4/3/2017	4/5/2017	First Class	SA-20830	Sue Ann Reed	Consumer	United States	Chicago	Illinois	60653	Central	OFF-AR-1	Office Supp	Art	Prang Colored F	\$7.06	3	\$0.20	\$2.21
27	7614	US-2015-	2/8/2015	2/11/2015	First Class	BH-11710	Brosina Hoffma	Consumer	United States	Garden City	Kansas	67846	Central	OFF-AR-1	Office Supp	Art	Sanford Colorif	\$5.76	2	\$0.00	\$1.73
28	6387	US-2017-	1/15/2017	1/18/2017	First Class	TB-21250	Tim Brockman	Consumer	United States	Austin	Texas	78745	Central	OFF-BI-10	Office Supp	Binders	Acco D-Ring Bin	\$4.28	1	\$0.80	-\$6.63
29	7787	US-2016-	5/17/2016	5/20/2016	First Class	LW-17215	Luke Weiss	Consumer	United States	Chicago	Illinois	60653	Central	OFF-BI-10	Office Supp	Binders	Acco Recycled 2	\$2.89	1	\$0.80	-\$4.77
30	4070	CA-2016-	8/28/2016	8/31/2016	First Class	TP-21415	Tom Prescott	Consumer	United States	Dallas	Texas	75081	Central	OFF-BI-10	Office Supp	Binders	Angle-D Binders	\$13.14	9	\$0.80	-\$21.68
31	8034	CA-2015-	6/25/2015	6/28/2015	First Class	MV-1748C	Mark Van Huff	Consumer	United States	Houston	Texas	77041	Central	OFF-BI-10	Office Supp	Binders	Avery Triangle Sl	\$0.98	2	\$0.80	-\$1.48
32	902	CA-2017-	11/11/2017	11/13/2017	First Class	TD-20995	Tamara Dahlen	Consumer	United States	Oakland	Texas	75043	Central	OFF-BI-10	Office Supp	Binders	Deluxe Heavy-D	\$18.34	4	\$0.80	\$32.09
33	5311	CA-2017-	11/19/2017	11/21/2017	First Class	NC-18415	Nathan Cano	Consumer	United States	Houston	Texas	77095	Central	OFF-BI-10	Office Supp	Binders	Fellowes 9560C	\$1,625.19	6	\$0.80	#####
34	1326	US-2014-	5/27/2014	5/30/2014	First Class	LE-18810	Laurel Elliston	Consumer	United States	Chicago	Illinois	60653	Central	OFF-BI-10	Office Supp	Binders	Fellowes Preser	\$17.46	6	\$0.80	\$30.56
35	1897	CA-2017-	10/3/2017	10/6/2017	First Class	AC-10450	Amy Cox	Consumer	United States	Minneapolis	Minnesota	55407	Central	OFF-BI-10	Office Supp	Binders	GBC DocuBind	\$1,793.98	2	\$0.00	\$843.17
36	3295	CA-2014-	8/15/2014	8/18/2014	First Class	KH-16510	Keith Herrera	Consumer	United States	Dallas	Texas	75081	Central	OFF-BI-10	Office Supp	Binders	GBC Durable Pl	\$30.96	8	\$0.80	-\$52.63
SuperstoreDataClean																					
SummaryCalculationsByRegion				SummaryCalculationsByCategory				SummaryCalcula													

Q: Part B: Descriptive Analytics Use Excel functions (e.g., `AVERAGE`, `COUNTIF`, `SUMIF`, `VLOOKUP`, pivot tables).

1. Calculate total sales, profit, and quantity by: • Region • Category • Customer Segment 2.

Summary Calculation by Region							
Region	Total Sales	Average Sales	Total Profit	Average Profit	Total Quantity	Average Quantity	Count of orders
Central	\$501,239.89	\$215.77	\$39,706.36	\$17.09	8780	3.78	2323
East	\$678,781.24	\$238.34	\$91,522.78	\$32.14	10618	3.73	2848
South	\$391,721.91	\$241.80	\$46,749.43	\$28.86	6209	3.83	1620
West	\$725,457.82	\$226.49	\$108,418.45	\$33.85	12266	3.83	3203

Summary Calculation by region (SUMIF, AVERAGEIF, COUNTIF)

Analytics for Decision Making

Summary Calculation by Category((SUMIF, AVERAGEIF, COUNTIF)

Summary Calculation by Category						
Total Sales	Average Sales	Total Profit	Average Profit	Total Quantity	Average Quantity	Count of orders
\$741,999.80	\$349.83	\$18,451.27	\$8.70	8028	3.79	2121
\$719,047.03	\$119.32	\$122,490.80	\$20.33	22906	3.80	6026
\$836,154.03	\$452.71	\$145,454.95	\$78.75	6939	3.76	1847

Summary Calculation by Customer Segment(SUMIF, AVERAGEIF, COUNTIF)

Summary Calculation by Segment							
Region	Total Sales	Average Sales	Total Profit	Average Profit	Total Quantity	Average Quantity	Count of orders
Consumer	\$1,161,401.34	\$223.73	\$134,119.21	\$25.84	19521	3.76	5191
Corporate	\$706,146.37	\$233.82	\$91,979.13	\$30.46	11608	3.84	3020
Home Office	\$429,653.15	\$240.97	\$60,298.68	\$33.82	6744	3.78	1783

Q: Use PivotTables to: Show sales performance by month and region.

Pivot Table: Show sales performance by Month and Region
 - Change the currency of sales

Monthly Sales by Region						
Sum of Sales	Column Labels					
Row Labels	Central	East	South	West	Grand Total	
2014	\$103,838.16	\$128,680.46	\$103,845.84	\$147,883.03	\$484,247.50	
2015	\$102,874.22	\$156,332.06	\$71,359.98	\$139,966.25	\$470,532.51	
2016	\$147,429.38	\$180,685.82	\$93,610.22	\$187,480.18	\$609,205.60	
2017	\$147,098.13	\$213,082.90	\$122,905.86	\$250,128.37	\$733,215.26	
Grand Total	\$501,239.89	\$678,781.24	\$391,721.91	\$725,457.82	\$2,297,200.86	

Pivot Table: Sales performance by Month and Region was analyzed. Sales were formatted as currency, with a total of **\$2,297,200.86** across all four regions (West, Central, East, South). The highest sales were recorded in the West region, **sales at \$250,128.37** in year 2017, while the lowest sales occurred in the South in year 2015 with sales **\$71,359.98**.

Analytics for Decision Making

Pivot Table: Rank the top 10 customers in sales. (Largest to Smallest)

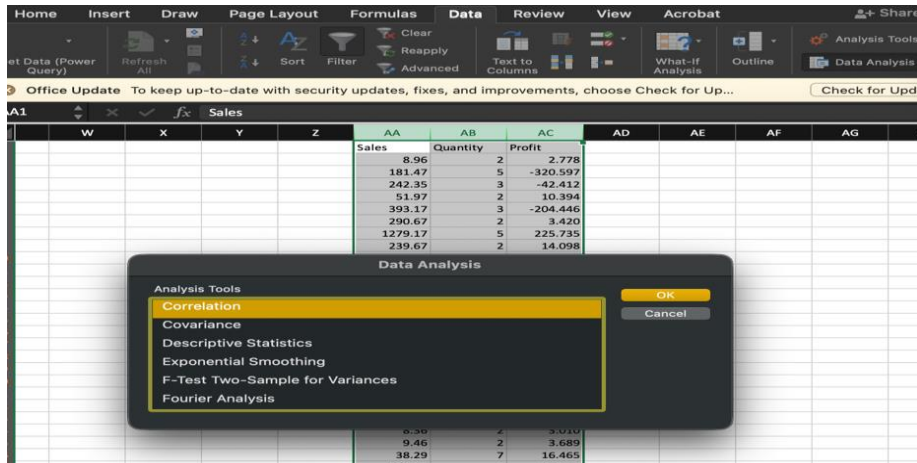
The customer **Sean Miller** ranks highest among the **top 10 sales customers** with sales of **\$25,043.05**, while the lowest among the top 10 is **Christopher Conant** with sales at **\$12,129.07**.

Row Labels	Sum of Sales
Sean Miller	\$25,043.05
Tamara Chand	\$19,052.22
Raymond Buch	\$15,117.34
Tom Ashbrook	\$14,595.62
Adrian Barton	\$14,473.57
Ken Lonsdale	\$14,175.23
Sanjit Chand	\$14,142.33
Hunter Lopez	\$12,873.30
Sanjit Engle	\$12,209.44
Christopher Conant	\$12,129.07

The screenshot displays the Microsoft Excel interface. A PivotTable is visible in the background, showing a list of customer names and their corresponding sales amounts. The context menu is open over the PivotTable, showing options such as 'Sort', 'Filter', 'Refresh', 'Remove "Sum of Sales"', 'Value Field Settings...', 'PivotTable Options...', and 'Hide Field List'. The 'Sort' option is selected, and a sub-menu is visible showing 'Sort Smallest to Largest' and 'Sort Largest to Smallest'. The 'Sort Largest to Smallest' option is highlighted. The PivotTable Fields task pane is also visible on the right side of the screen, showing the list of fields and the current PivotTable structure.

Analytics for Decision Making

Q: Part C: Inferential Analytics Use Excel's Data Analysis ToolPak to perform:



Correlation Analysis: • Examine the correlation between Sales, Profit, and Quantity
The result is :

Corelation :			
	<i>Sales</i>	<i>Quantity</i>	<i>Profit</i>
<i>Sales</i>	1	0.201	0.482
<i>Quantity</i>	0.215	1	0.067
<i>Profit</i>	0.647	0.09	1

Interpret strength and direction

The diagonal (Sales-Sales, Quantity-Quantity, Profit-Profit) is always **1**, because a variable is perfectly correlated with itself.

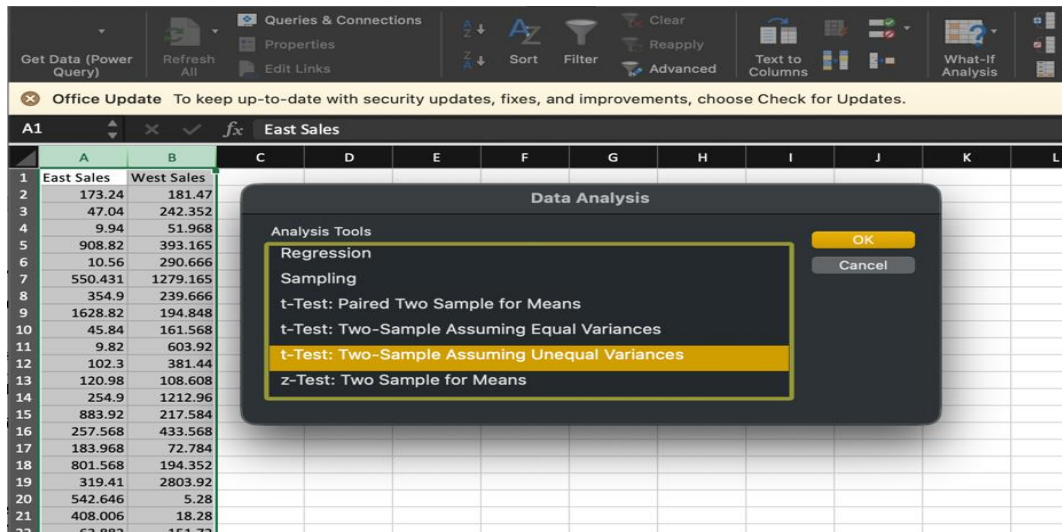
Every other cell represents the correlation between **two different variables**:

- Sales & Profit (0.482):** Moderate positive → increasing sales tends to increase profit.
- Sales & Quantity (0.201):** Weak positive → more units sold slightly increases sales.
- Quantity & Profit (0.067):** Negligible → quantity sold doesn't strongly affect profit.

All correlations are positive, but only **Sales & Profit** is meaningful.

Analytics for Decision Making

2. Hypothesis Testing: Question: Is there a significant difference in average sales between the East and West regions? Prepare Data: Create two separates• Conduct two-sample t-test (assume unequal variances) State null/alternative hypotheses, significance level ($\alpha = 0.05$), and interpret the result.



Hypothesis Testing (Two-Sample t-Test)

The output is:

F-Test Two-Sample for Variances		
	East Sales	West Sales
Mean	238.3361096	226.4932328
Variance	385284.1966	275495.7363
Observations	2848	3203
df	2847	3202
F	1.398512376	
P(F<=f) one-tail	1.54614E-20	
F Critical one-tail	1.061697947	

Null Hypothesis (H_0): **There is no significant difference in the average sales between the East and West regions** ($\mu_{\text{East}} = \mu_{\text{West}}$).

Alternative Hypothesis (H_a): **There is a significant difference in the average sales between the East and West regions** ($\mu_{\text{East}} \neq \mu_{\text{West}}$).

Analytics for Decision Making

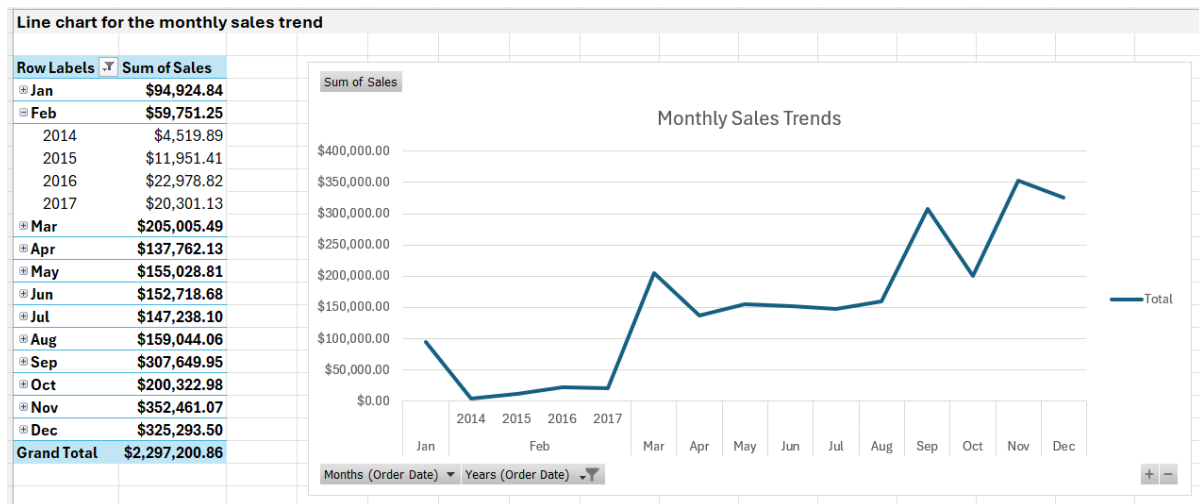
Q: Part D: Data Visualization

Using charts and dashboards in Excel:

1 Create the following visualizations:

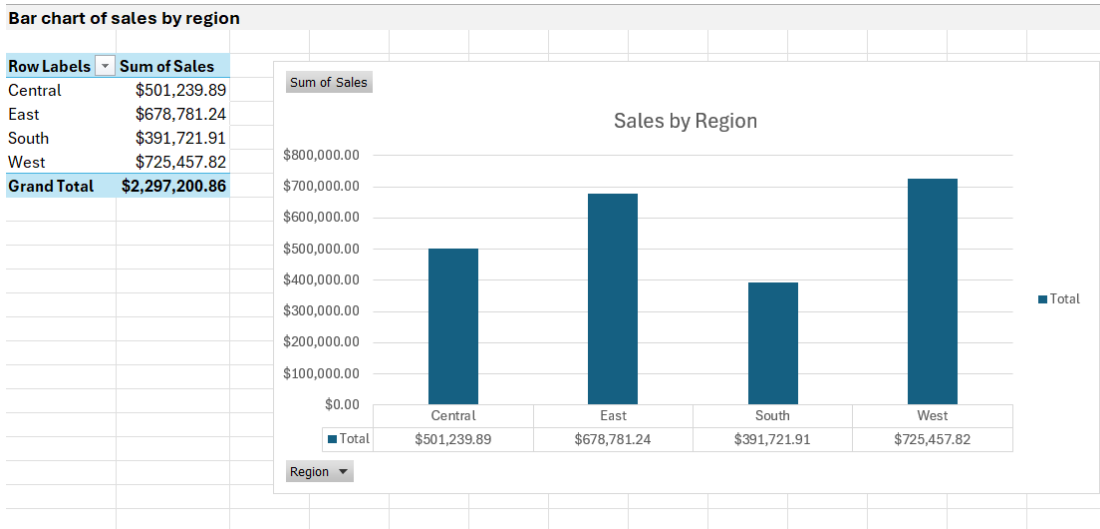
1. Line chart for the monthly sales trend

The chart displays monthly sales by year. For example, in February 2017, total sales increased to 20,301.13, showing a clear rise compared to the same month in 2014.



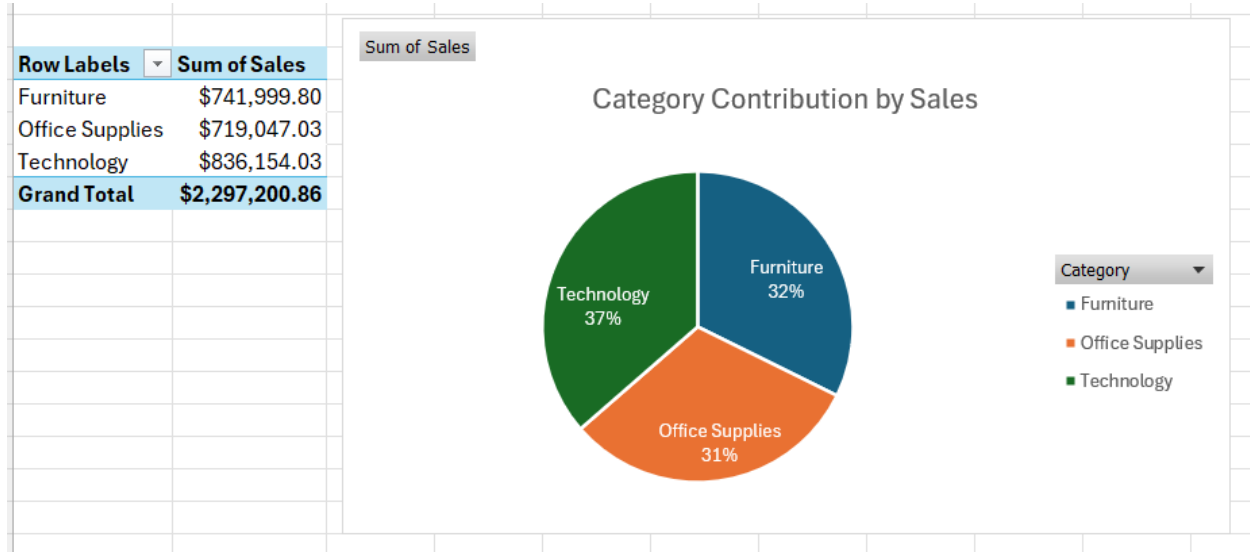
2. Bar chart of sales by region

The West region recorded sales at \$725,445.82 compared to the other regions.



3. Pie chart of category contribution to total sales

Technology contributed **32%** of total sales, slightly higher than Office Supplies at **31%**. The Furniture and Office Supplies categories differed by less than **1%** in their overall sales contribution.



Heatmap (conditional formatting) for profit margins across regions and categories

A heatmap using conditional formatting was created to display profit margins across regions and categories. The West region had the highest average profit margin of 28.58% in Office Supplies, followed by the East with the same category. The Central region showed a negative margin of -15.89% of Office Supplies.

Analytics for Decision Making

Heatmap (conditional formatting) for profit margins across regions and categories

Average of Profit Margin	Column Labels			
Row Labels	Furniture	Office Supplies	Technology	Grand Total
Central	-18.58	-15.89	17.51	-10.41
East	9.24	20.63	12.62	16.72
South	14.17	16.32	18.95	16.35
West	9.77	28.58	15.32	21.95
Grand Total	\$ 3.88	\$ 13.80	\$ 15.61	\$ 12.03

Q: Build a dashboard in a separate Excel sheet using slicers for:

- Region
- Product Category



Part E: Insight Generation & Recommendation

Write a one-page summary based on the evidence from Parts B, C, and D.

Summary Report:

1. Top Three Insights are as follows:

- **Insight 1 (Descriptive Profitability):** Technology products generate the highest total sales, but the Furniture category consistently reports the lowest profit margin, indicating potential pricing or cost issues.
- **Insight 2 (Trend/Anomaly):** Monthly sales show a clear seasonal spike in the last quarter (October-December) across all regions, suggesting this is the key time for resource allocation and marketing efforts.
- **Insight 3 (Inferential - T-test/Correlation):** Despite regional sales differences, the t-test revealed there is *no* statistically significant difference in the average sales value between the East and West regions ($\text{p-value} > 0.05$), suggesting business strategies should not be based solely on the overall sales total.

2. Focus Areas for Growth

- The **West** region accounts for the largest share of sales, while the **South** underperforms relative to its market potential

3. Anomalies / Risks:

- Certain categories/regions have negative or very low profits despite high sales — potential pricing or cost issues.

4. Actionable Recommendation

A specific, measurable action based on an insight.

- Conduct a deeper unit-cost analysis for the **Furniture** category in the **Central** region, specifically examining the 'Tables' sub-category, which appears in the lowest tier of the profit heatmap. Implement a minimum discount floor to prevent margin erosion from excessive price competition." Promote high-margin products in underperforming regions.
- Reduce discounts on low-margin, high-volume items.
- Monitor performance regularly using the dashboard with slicers.